

Problem E. Simple Stack

Time Limit 2000 ms

Mem Limit 131072 kB

Design and implement the data structure **stack**. Write a program that describes the stack and simulates stack operations, implementing all the methods specified below. The program reads a sequence of commands and, depending on the command, performs the corresponding operation. After executing each command, the program must print one line of output. The possible commands are:

- **push** n — Add the number n to the stack (the value n is given after the command). Print **ok**.
- **pop** — Remove the last element from the stack. Print its value.
- **back** — Print the value of the last element without removing it from the stack.
- **size** — Print the number of elements in the stack.
- **clear** — Clear the stack and print **ok**.
- **exit** — Print **bye** and terminate the program.

Stack
-m: int*
-_size: int
+Stack(_size: int = 110)
+push(v: int)
+pop(): int
+back(): int
+size(): int
+clear()
+exit()

It is guaranteed that the sequence of input commands satisfies the following requirements: the maximum number of elements in the stack at any time does not exceed 100, and all **pop** and **back** commands are correct, that is, at the moment they are performed, the stack contains at least one element.

Input

Each line contains a single command.

Output

For each command, output the corresponding result on a separate line.

Examples

Input	Output
push 2	ok
push 3	ok
push 5	ok
back	5
size	3
pop	5
size	2
push 7	ok
pop	7
clear	ok
size	0
exit	bye